Notes: Barberis, Boycko, Shleifer, and Tsukanova (JPE, 1996)

How Does Privatization Work? Evidence from the Russian Shops

Contents

1 Big picture

- **Question.** How does privatization improve firm performance? Does it work mainly by giving old insiders stronger equity incentives, or by bringing in new owners and managers with market-oriented human capital?
- **Empirical setting.** The paper studies a survey of Russian retail shops privatized during the early 1990s transition. The setting is useful because many shops moved quickly from state ownership to different private ownership structures, creating variation in owners, managers, and incentives.
- **Core outcomes.** The authors measure restructuring using four concrete actions: major renovation, supplier change, longer opening hours, and layoffs.
- **Main result.** Restructuring is strongly associated with new owners and new managers. Shops with complete ownership change or management layoffs are more likely to renovate, switch suppliers, extend hours, and restructure employment.
- **Negative result.** Equity incentives for old managers do not appear to be the main force. Management ownership and outside investor ownership have weak, unstable, or insignificant effects once new human capital is separated from cash-flow incentives.
- **Mechanism.** The evidence points to selection and human capital: privatization works when it reallocates control to people who have the skills and motivation to operate in a market economy.
- **Methodological contribution.** Instead of comparing private and state firms only at the outcome level, the paper decomposes privatization into alternative mechanisms: new owners, new managers, managerial equity, outside equity, privatization method, and property rights over premises.
- **Economic takeaway.** Ownership reform is not just about changing residual claims. Transition economies also require replacing or disciplining human capital that was selected for political and bureaucratic objectives rather than commercial performance.

2 Incremental contribution of this paper

2.1 Contribution relative to the privatization literature

Earlier privatization studies had established that private firms are often more efficient than state firms and that privatized firms often improve performance. However, those studies generally did not identify *why* privatization works. Barberis, Boycko, Shleifer, and Tsukanova make three incremental contributions.

1. **They separate privatization mechanisms.** The paper distinguishes between equity incentives for old insiders and the arrival of new human capital through new owners and managers.
2. **They use micro-level restructuring outcomes.** Rather than relying on accounting performance alone, they study direct operational changes: renovation, supplier changes, longer hours, and layoffs.
3. **They exploit institutional details of Russian privatization.** Variation in auction/competition sales, shop premises ownership, and management turnover helps identify which features of privatization matter.

Interpretation: The incremental insight is that privatization is not a black-box transfer from state to private ownership. Its effect depends on who receives control and whether those people bring market-oriented skills.

2.2 Contribution relative to agency and incentive theories

A standard agency view predicts that managers who own more equity should restructure more because they internalize more of the gains from profit improvement. The paper tests this prediction directly using management ownership and outside investor ownership.

- If equity incentives are the main mechanism, higher management ownership should predict restructuring.
- If new human capital is the main mechanism, complete ownership change and management layoffs should predict restructuring.
- The data favor the second mechanism.

Interpretation: The paper does not deny the importance of incentives in general. Its contribution is to show that in the Russian shop context, the binding problem was not primarily weak effort by old managers but the need to replace or supplement old human capital.

2.3 Contribution relative to transition economics

The transition from central planning to markets required more than legal privatization. Firms needed new suppliers, new operating hours, new renovation investment, and new labor decisions. The paper provides early evidence that these changes were more likely when privatization changed the people in charge.

Interpretation: The paper shifts attention from ownership labels to control rights and human capital. It argues that successful transition depends on the reallocation of control to individuals capable of acting on market opportunities.

3 Data and measurement

3.1 Survey sample

- The paper uses a survey of 452 Russian shops, most privatized between 1992 and 1993.
- The sample is composed of retail stores, which are attractive for measurement because restructuring actions are observable and relatively quick.
- Because responses are incomplete for some variables, the effective samples differ across outcomes.
- The main regression samples are approximately 331–340 observations for renovation, supplier change, and longer hours, and 266 observations for layoffs.

Interpretation: Retail shops are not representative of all privatized firms, but they are ideal for studying early restructuring because changes in operations can happen quickly after privatization.

3.2 Restructuring outcomes

The paper studies four restructuring measures:

1. **Renovation:** equals one if a major capital renovation took place.
2. **Supplier change:** equals one if more than 50 percent of suppliers changed.
3. **Longer hours:** equals one if the shop increased its hours of operation.
4. **Employee layoffs:** equals one if layoffs occurred.

Interpretation: These outcomes capture both investment restructuring and operational restructuring. Renovation and supplier changes indicate real changes in business strategy; longer hours and layoffs capture changes in labor and service intensity.

3.3 Human capital variables

The key human capital variables are:

- **Complete ownership change:** equals one if 100 percent of the owners are new to the firm.
- **Management layoffs:** equals one if managers were laid off.

Interpretation: These variables proxy for the arrival of new people. Complete ownership change brings new owners; management layoffs indicate that old managers were replaced or disciplined.

3.4 Cash-flow incentive variables

The paper measures cash-flow incentives through ownership shares:

- **Management ownership:** the percentage of the shop owned by management, whether old or new.
- **Outside investor ownership:** the percentage of the shop owned by outsiders, whether individuals or legal entities.

Interpretation: These variables proxy for financial incentives. If equity ownership is the main channel, these variables should strongly predict restructuring even when human capital change is absent.

3.5 Privatization and property-rights variables

Additional variables include:

- **Date:** number of months after June 1992 that privatization occurred.
- **Shop owns its premises:** equals one if the shop owns its physical/legal premises.
- **Auction dummy:** equals one if the shop was sold by auction.
- **Competition dummy:** equals one if the shop was sold in a competition.
- **Layoff restrictions:** equals one if restrictions on employee layoffs were reported.

Interpretation: These variables help separate privatization method and property rights from ownership incentives. They also serve as instruments in the two-stage least squares analysis.

4 Models and empirical specifications

4.1 Baseline linear probability model

For each restructuring outcome Y_i, the paper estimates linear probability models of the form

$$Y_i = \alpha + \beta_1 Date_i + \beta_2 CompleteOwnership_i + \beta_3 ManagementLayoff_i + \varepsilon_i. \tag{1}$$

Extended specifications add incentive variables:

$$Y_i = \alpha + \beta_1 Date_i + \beta_2 HumanCapital_i + \beta_3 Incentives_i + \beta_4 LayoffRestrictions_i + \varepsilon_i. \tag{2}$$

Interpretation: The coefficients are changes in the probability of restructuring. For example, a coefficient of 0.20 means a 20 percentage point increase in the probability of restructuring.

4.2 Human capital specification

The human capital regressions focus on complete ownership change and management layoffs:

$$Y_i = \alpha + \beta_1 Date_i + \beta_2 CompleteOwnership_i + \beta_3 ManagementLayoff_i + \beta_4 CompleteOwnership_i \times ManagementLayoff_i + \gamma_i. \tag{3}$$

Interpretation: This specification asks whether restructuring is more likely when owners and managers are new. The interaction tests whether new owners and new managers are complements.

4.3 Cash-flow incentive specification

The incentive regressions replace human capital variables with ownership shares:

$$Y_i = \alpha + \beta_1 Date_i + \beta_2 ManagementOwnership_i + \beta_3 OutsideOwnership_i + \beta_4 LayoffRestrictions_i + \varepsilon_i. \tag{4}$$

Interpretation: This specification asks whether the share of cash-flow rights held by management or outside investors predicts restructuring.

4.4 Subsample test for old human capital

To isolate incentives for old managers, the paper studies a subsample with:

- no complete ownership change;
- no management layoffs;
- no major new human capital entering control.

The regression is

$$Y_i = \alpha + \beta_1 Date_i + \beta_2 ManagementOwnership_i + \beta_3 LayoffRestrictions_i + \varepsilon_i. \tag{5}$$

Interpretation: If equity incentives for old managers are powerful, management ownership should be positive in this restricted sample. Table 4 and Table 8 show that it is not.

4.5 Two-stage least squares design

The first stage predicts ownership and human capital variables using privatization method and property rights:

$$HumanCapital_i \text{ or } Incentive_i = \pi_0 + \pi_1 Date_i + \pi_2 Premises_i + \pi_3 Auction_i + \pi_4 Competition_i + u_i. \tag{6}$$

The second stage uses fitted values:

$$Y_i = \alpha + \beta_1 \widehat{HumanCapital}_i + \beta_2 \widehat{Incentive}_i + \gamma Controls_i + \varepsilon_i. \tag{7}$$

Interpretation: The IV design uses exogenous-looking features of privatization implementation to address selection into ownership and incentive structures. The identification assumption is that privatization method and premises ownership affect restructuring through the human capital and incentive variables, not directly through unobserved restructuring demand.

5 Identification and empirical design

5.1 What identifies the main effect

- The paper exploits cross-shop variation in whether ownership changed completely, whether managers were laid off, and who owned equity after privatization.
- It controls for privatization date and layoff restrictions.
- It uses privatization method and premises ownership as instruments for the human capital and ownership variables.

Interpretation: The core identifying contrast is between shops with different post-privatization control structures but similar privatization timing and institutional setting.

5.2 Why privatization method matters

Auctions and competitions were more likely to bring in new owners and outside investors, while shops transferred to insiders retained more old human capital. This institutional variation helps distinguish the role of new people from the role of residual claims.

Interpretation: The privatization method provides variation in who ends up controlling the shop. This is what allows the paper to ask whether restructuring comes from incentives or from people.

5.3 Threats to identification

- Shops selected into auctions or competitions may have differed before privatization.
- New owners may have chosen shops with better restructuring opportunities.
- Survey variables may be measured with error.
- The sample is limited to retail shops and may not generalize to industrial firms.
- IV validity requires that instruments do not directly affect restructuring except through human capital and incentives.

Interpretation: The paper is strongest as early micro evidence on privatization mechanisms. It is less strong as a clean quasi-experiment because privatization methods were not randomly assigned.

5.4 Why the evidence still matters

Even with these limitations, the results are informative because the pattern is consistent across many tests: new owners and managers predict restructuring, while old-management equity incentives do not.

Interpretation: The empirical design is not perfect, but the contrast between human capital and incentive channels is unusually direct for the privatization literature.

6 Tables and figures

6.1 Table 1: Probability of restructuring

Read:

- The unconditional probability of restructuring is 13.6 percent for renovation, 45.2 percent for supplier change, 16.5 percent for longer hours, and 46.2 percent for employee layoffs.
- Shops with complete ownership change are more likely to renovate and change suppliers.
- Shops with management layoffs have much higher probabilities of renovation, supplier change, longer hours, and employee layoffs.
- Higher management ownership is associated with more layoffs but does not uniformly predict other restructuring actions.
- Competitive sale methods are associated with supplier changes but not uniformly with all outcomes.

Economic message: Descriptively, new owners and new managers are much more closely related to restructuring than equity shares alone.

TABLE 1

PROBABILITY OF RESTRUCTURING

Variable	Renovation (N = 331)	Supplier Change (N = 336)	Longer Hours (N = 334)	Employee Layoffs (N = 266)
Unrestricted mean	.1360	.4524	.1647	.4624
Complete ownership change:				
No	.1026	.3849	.1555	.4639
Yes	.2165	.6186	.1875	.4583
Management layoffs:				
No	.1074	.4066	.125	.3973
Yes	.2623	.6508	.3387	.766
Management ownership:				
<23%	.1091	.4881	.1446	.3511
>23%	.1627	.4167	.1845	.5701
Outside investor ownership:				
=0	.1141	.369	.1658	.549
>0	.1633	.557	.1633	.3451
Shop owns its premises:				
No	.1131	.4533	.1518	.5054
Yes	.1818	.4505	.1909	.3625
Competitive sale method:				
No	.1197	.3058	.1441	.5169
Yes	.1355	.5209	.1745	.4206

NOTE.—Unconditional and conditional means of four measures of restructuring: renovation, one if capital renovation was done, and zero otherwise; supplier change, one if more than 50 percent of the suppliers were changed, zero otherwise; longer hours, one if longer hours were worked, zero otherwise; and employee layoffs, one if layoffs were made, zero otherwise. Complete ownership change is one if 100 percent of the owners are new to the firm, zero otherwise. Management layoffs is one if managers were laid off, zero otherwise. Management ownership is the percentage of the shop owned by the management, whether old or new. Outside investor ownership is the percentage of the shop owned by outsiders, whether physical or legal entities. A shop is sold in a competitive sale

TABLE 2

RESTRUCTURING AS A FUNCTION OF HUMAN CAPITAL CHANGE

	Renovation (N = 331)			Supplier Change (N = 336)			Longer Hours (N = 334)			Employee Layoffs (N = 266)		
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Constant	.265	.255	.238	.601	.597	.560	.317	.249	.26	.774	.641	.65
	(.067)	(.071)	(.070)	(.079)	(.082)	(.083)	(.067)	(.065)	(.066)	(.095)	(.098)	(.099)
Date	−.016	−.014	−.014	−.022	−.019	−.019	−.016	−.012	−.012	−.020	−.014	−.014
	(.006)	(.006)	(.006)	(.007)	(.007)	(.007)	(.006)	(.006)	(.005)	(.009)	(.008)	(.008)
Complete ownership change	.104		.071	.223		.178	.024		−.033	.025		−.093
	(.045)		(.051)	(.058)		(.071)	(.046)		(.045)	(.068)		(.080)
Management layoffs		.13	.087		.210	.125		.192	.2		.322	.329
		(.060)	(.088)		(.068)	(.103)		(.064)	(.099)		(.074)	(.102)
Complete ownership change × management layoff			.028			.033			.009			.05
			(.124)			(.139)			(.130)			(.149)
Layoff restrictions										−.172	−.142	−.122
										(.065)	(.063)	(.065)
Adjusted R^2	4.31	4.49	4.88	6.24	4.73	6.78	1.85	5.69	5.25	4.14	9.96	9.74

NOTE.—OLS regression estimates of the probability of four measures of restructuring as a function of the date since privatization and variables indicating human capital change. The measures of restructuring are renovation, one if capital renovation was done, and zero otherwise; supplier change, one if more than 50 percent of the suppliers were changed, zero otherwise; longer hours, one if longer hours were worked, zero otherwise; and employee layoffs, one if layoffs were made, zero otherwise. Date is the number of months after June 1992 that privatization occurred. Complete ownership change is one if 100 percent of the owners are new to the firm, zero otherwise. Management layoffs is one if managers were laid off, zero otherwise. The employee layoff regressions also control for layoff restrictions, one if restrictions were reported, zero otherwise. Heteroskedasticity-consistent standard errors are in parentheses.

6.2 Table 2: Restructuring as a function of human capital change

Read:

- Complete ownership change is positively related to renovation and supplier changes.
- Management layoffs are positively related to renovation, supplier change, longer hours, and layoffs.
- The interaction between ownership change and management layoffs is generally not the main driver.
- Layoff restrictions reduce the probability of employee layoffs, as expected.
- Date controls indicate that restructuring evolves over time after privatization.

Economic message: New owners and managers drive restructuring. Privatization works when it brings in people capable of changing operations.

6.3 Table 3: Restructuring as a function of cash-flow incentives

Read:

- Management ownership coefficients are close to zero and generally insignificant.
- Outside investor ownership coefficients are also weak and not robust.
- Cash-flow incentives do not explain renovation, supplier changes, or longer hours nearly as strongly as human capital variables do.
- Layoff restrictions matter for layoffs, not for the other forms of restructuring.

Economic message: The equity-incentive channel is weak in this setting. Ownership claims alone do not make old managers restructure.

TABLE 3

RESTRUCTURING AS A FUNCTION OF CASH FLOW INCENTIVES

	Renovation (N = 353)			Supplier Change (N = 340)			Longer Hours (N = 338)			Employee Layoffs (N = 266)		
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Constant	.393	.362	.304	.719	.607	.474	.257	.324	.151	.671	.786	.707
	(.076)	(.069)	(.090)	(.082)	(.077)	(.101)	(.066)	(.065)	(.077)	(.110)	(.093)	(.123)
Date	−.023	−.023	−.022	−.023	−.024	−.023	−.015	−.016	−.014	−.019	−.019	−.018
	(.006)	(.006)	(.006)	(.007)	(.007)	(.007)	(.005)	(.006)	(.005)	(.009)	(.008)	(.008)
Management ownership	−.000		.001	−.001		.003	.002		.003	.002		.002
	(.001)		(.001)	(.001)		(.001)	(.001)		(.001)	(.001)		(.002)
Outside investor ownership		.001	.001		.002	.003		−.000	.001		−.001	−.001
		(.000)	(.001)		(.001)	(.001)		(.000)	(.000)		(.001)	(.001)
Layoff restrictions										−.127	−.131	−.121
										(.068)	(.068)	(.068)
Adjusted R^2	4.34	4.99	5.06	2.41	5.74	6.48	3.43	1.75	4.66	5.39	5.20	5.18

NOTE.—OLS regression estimates of the probability of four measures of restructuring as a function of the date since privatization and variables measuring cash flow incentives. The measures of restructuring are renovation, one if capital renovation was done, and zero otherwise; supplier change, one if more than 50 percent of the suppliers were changed, zero otherwise; longer hours, one if longer hours were worked, zero otherwise; and employee layoffs, one if layoffs were made, zero otherwise. Date is the number of months after June 1992 that privatization occurred. Management ownership is the percentage of the shop owned by the management, whether old or new. Outside investor ownership is the percentage of the shop owned by outsiders, whether physical or legal entities. The employee layoffs regressions also control for layoff restrictions, one if restrictions were reported, zero otherwise. Heteroskedasticity-consistent standard errors are in parentheses.

TABLE 4

RESTRUCTURING AS A FUNCTION OF MANAGEMENT OWNERSHIP

	Subsample: No Complete Ownership Change and No Management Layoffs			
Variable	Renovation (N = 172)	Supplier Change (N = 174)	Longer Hours (N = 174)	Employee Layoffs (N = 143)
Constant	.263	.456	1.866	.472
	(.095)	(.132)	(.088)	(.188)
Date	−.016	−.014	.008	−.019
	(.007)	(.010)	(.006)	(.012)
Management ownership	−.000	.001	−.002	.001
	(.001)	(.002)	(.001)	(.002)
Layoff restrictions				.117
				(.085)
Adjusted R^2	2.44	.14	1.30	1.83

NOTE.—OLS regression estimates of the probability of four measures of restructuring as a function of the date since privatization and management ownership, a variable measuring cash flow incentives, within a subsample in which no managers were laid off and people new to the shop own less than 50 percent of the shop. The measures of restructuring are renovation, one if capital renovation was done, and zero otherwise; supplier change, one if more than 50 percent of the suppliers were changed, zero otherwise; longer hours, one if longer hours were worked, zero otherwise; and employee layoffs, one if layoffs were made, zero otherwise. Date is the number of months after June 1992 that privatization occurred. Management ownership is the percentage of the shop owned by the management, whether old or new. The employee layoffs regressions also control for layoff restrictions, one [illegible]

6.4 Table 4: Management ownership in a restricted subsample

Read:

- The sample excludes shops with complete ownership change and management layoffs.
- In this sample, management ownership does not significantly predict renovation, supplier change, longer hours, or layoffs.
- The point estimates are tiny and sometimes have the wrong sign.

Economic message: This is a sharp test of incentives for old human capital. The result says that giving old managers equity did not by itself produce restructuring.

6.5 Table 5: First-stage instruments

Read:

- Auction and competition sales strongly predict complete ownership change and outside investor ownership.
- Auction and competition sales reduce management ownership.
- Shop ownership of premises predicts some ownership outcomes and helps identify property-rights-related variation.
- First-stage R^2 values are reasonably high for ownership change and outside investor ownership.

Economic message: Privatization method provides systematic variation in who controls the shop. This supports the IV strategy used in the next tables.

TABLE 5

PREDICTING HUMAN CAPITAL CHANGE AND OWNERSHIP

Variable Being Instrumented	Complete Ownership Change (N = 327)	Management Layoffs (N = 327)	Management Ownership (N = 331)	Outside Investor Ownership (N = 331)
Constant	.108	.264	47.144	1.208
	(.058)	(.074)	(4.432)	(5.486)
Date	−.013	−.021	−.124	−.198
	(.006)	(.006)	(.372)	(.521)
Shop owns its premises	.109	.050	−9.166	15.156
	(.048)	(.045)	(2.797)	(4.332)
Auction dummy	.488	.266	−20.726	51.336
	(.056)	(.058)	(3.933)	(5.256)
Competition dummy	.387	.108	−24.241	49.500
	(.045)	(.041)	(2.745)	(4.264)
Adjusted R^2	23	10.9	21.1	32.5

NOTE.—First-stage results from a two-stage least squares procedure in which the variables measuring human capital change and the cash flow incentives are regressed on four instrumental variables. Complete ownership change is one if 100 percent of the owners are new to the firm, zero otherwise. Management layoffs is one if managers were laid off, zero otherwise. Management ownership is the percentage of the shop owned by the management, whether old or new. Outside investor ownership is the percentage of the shop owned by outsiders, whether physical or legal entities. The instrumental variables are date, the number of months since June 1992 that

TABLE 6

TWO-STAGE LEAST SQUARES ESTIMATES OF THE EFFECTS OF HUMAN CAPITAL CHANGE

	Renovation (N = 327)		Supplier Change (N = 332)		Longer Hours (N = 330)		Employee Layoffs (N = 265)	
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	.260	.164	.494	.206	.258	.088	.808	.928
	(.076)	(.108)	(.090)	(.161)	(.071)	(.107)	(.102)	(.174)
Date	−.017	−.010	−.021	.002	−.014	−.003	−.021	−.027
	(.006)	(.008)	(.008)	(.012)	(.006)	(.008)	(.009)	(.012)
Complete ownership change	.129		.557		.172		−.187	
	(.091)		(.116)		(.094)		(.181)	
Management layoffs		.360		1.158		.572		−.379
		(.194)		(.328)		(.213)		(.336)
Layoff restrictions							−.129	−.193
							(.074)	(.074)

NOTE.—Second-stage results of a two-stage least squares procedure in which four measures of restructuring are regressed on fitted values of the explanatory variables. The measures of restructuring are renovation, one if capital renovation was done, and zero otherwise; supplier change, one if more than 50 percent of the suppliers were changed, zero otherwise; longer hours, one if longer hours were worked, zero otherwise; and employee layoffs, one if layoffs were made, zero otherwise. The explanatory variables are date, the number of months after June 1992 that privatization occurred; complete ownership change, one if 100 percent of the owners are new to the firm, zero otherwise; and management layoffs, one if managers were laid off, zero otherwise. The employee layoffs regressions also control for layoff restrictions, one if restrictions were reported, zero otherwise. First-stage results are in table 5. Heteroskedasticity-consistent standard errors are in parentheses.

6.6 Table 6: Two-stage least squares estimates of human capital change

Read:

- The second-stage results use fitted values of complete ownership change and management layoffs.
- Complete ownership change remains positively related to supplier change and longer hours.
- Management layoffs remain positively related to renovation, supplier change, and longer hours.
- Effects on employee layoffs are less stable and sometimes negative.

Economic message: The IV results confirm the OLS evidence that new human capital is central to restructuring.

6.7 Table 7: Two-stage least squares estimates of incentives

Read:

- The second-stage estimates use fitted values of management ownership and outside investor ownership.
- Management ownership does not consistently predict restructuring.
- Outside investor ownership is not robustly positive across outcomes.
- The incentive results are weaker and less interpretable than the human-capital results.

Economic message: Even instrumented incentive variables do not deliver a strong restructuring effect. The main channel remains selection of new people, not old insiders' cash-flow claims.

6.8 Table 8: Two-stage estimates in the restricted subsample

Read:

- The table repeats the incentive test in shops with no complete ownership change and no management layoffs.
- Management ownership is not positively related to restructuring.
- The coefficients remain small and statistically weak.

TABLE 7

Two-Stage Least Squares Estimates of the Effects of Incentives

	Renovation (N = 331)			Supplier Change (N = 336)			Longer Hours (N = 334)			Employee Layoffs (N = 265)		
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Constant	.362	.272	−.225	.957	.538	−1.205	.370	.279	−1.171	.607	.777	−.775
	(.083)	(.071)	(.552)	(.110)	(.083)	(1.121)	(.087)	(.066)	(.878)	(.191)	(.096)	(2.239)
Date	−.018	−.018	−.015	−.028	−.026	−.016	−.016	−.016	−.008	−.017	−.018	−.013
	(.006)	(.006)	(.007)	(.008)	(.007)	(.015)	(.006)	(.006)	(.011)	(.009)	(.009)	(.016)
Management ownership	−.002		.010	−.008		.036	−.002		.030	.004		.032
	(.002)		(.011)	(.002)		(.023)	(.002)		(.018)	(.004)		(.045)
Outside investor ownership		.001	.006		.004	.022		.001	.015		−.001	.013
		(.001)	(.005)		(.001)	(.011)		(.000)	(.009)		(.002)	(.021)
Layoff restrictions										−.103	−.126	.027
										(.088)	(.079)	(.257)

Note.—Second-stage results of a two-stage least squares procedure in which four measures of restructuring are regressed on fitted values of the explanatory variables. The measures of restructuring are renovation, one if capital renovation was done, and zero otherwise; supplier change, one if more than 50 percent of the suppliers were changed, zero otherwise; longer hours, one if longer hours were worked, zero otherwise; and employee layoffs, one if layoffs were made, zero otherwise. The explanatory variables are date, the number of months after June 1992 that privatization occurred; management ownership, the percentage of the shop owned by the management, whether old or new; and outside investor ownership, the percentage of the shop owned by outsiders, whether physical or legal entities. The employee layoff regressions also control for layoff restrictions, one if restrictions were reported, zero otherwise. First-stage results are in table 5. Heteroskedasticity-consistent standard errors are in parentheses.

TABLE 8

Two-Stage Least Squares Estimates of the Effects of Incentives

	Subsample: No Complete Ownership Change and No Management Layoffs			
Variable	Renovation (N = 172)	Supplier Change (N = 174)	Longer Hours (N = 174)	Employee Layoffs (N = 143)
Constant	−.060	1.599	1.644	.647
	(.305)	(.612)	(.331)	(.544)
Date	−.010	−.032	.012	−.022
	(.008)	(.016)	(.008)	(.015)
Management ownership	.006	−.022	.003	−.004
	(.006)	(.012)	(.007)	(.014)
Layoff restrictions				.149
				(.128)

Note.—Second-stage results of a two-stage least squares procedure in which four measures of restructuring are regressed on fitted values of the explanatory variables, within the subsample in which no managers were laid off and people new to the shop own less than 50 percent of the shop. The measures of restructuring are renovation, one if capital renovation was done, and zero otherwise; supplier change, one if more than 50 percent of the suppliers were changed, zero otherwise; longer hours, one if longer hours were worked, zero otherwise; and employee layoffs, one if layoffs were made, zero otherwise. The explanatory variables are date, the number of months after June 1992 that privatization occurred, and management ownership, the percentage of the shop owned by the management, whether old or new. The employee layoff regressions also control for layoff restrictions, one if restrictions were reported, zero otherwise. Instruments are described in table 5. Heteroskedasticity-consistent standard errors are in parentheses.

Economic message: This table strengthens the conclusion that equity incentives for old managers did not drive restructuring in the Russian shops.

7 Short summary

- The paper studies 452 Russian shops privatized during the early transition period.
- It asks whether privatization works through equity incentives or through new human capital.
- Restructuring is measured by renovation, supplier change, longer hours, and layoffs.
- Complete ownership change and management layoffs are strongly associated with restructuring.
- Management ownership and outside investor ownership do not consistently predict restructuring.
- A restricted-sample analysis shows that old managers with more equity did not restructure more.
- IV results using privatization method and premises ownership support the human-capital interpretation.
- The paper's main contribution is to decompose privatization mechanisms rather than treat privatization as a single ownership dummy.

8 Conclusion

8.1 Core conclusion

Barberis, Boycko, Shleifer, and Tsukanova conclude that privatization worked in Russian shops primarily when it brought in new owners and new managers. New people, not simply new incentives for old people, were the key driver of restructuring.

8.2 Economic intuition

State-enterprise managers were selected for operating in a planned economy: maintaining political relationships, lobbying, and satisfying bureaucratic rules. A market economy requires different skills: finding suppliers, renovating stores, extending hours, and adjusting employment. Privatization is effective when it reallocates control to people with these market-oriented skills.

8.3 Incremental takeaway

The paper's incremental contribution is to show that the effect of privatization depends on the mechanism of control transfer. Private ownership is not enough if the same old human capital remains in place and only receives equity incentives. Effective privatization requires the reallocation of control to new human capital.